PUSAN NATIONAL UNIVERSITY

Dept. of Aerospace Engineering

Computer-Aided Engineering

Mid-exam. Oct. 27th, 2016

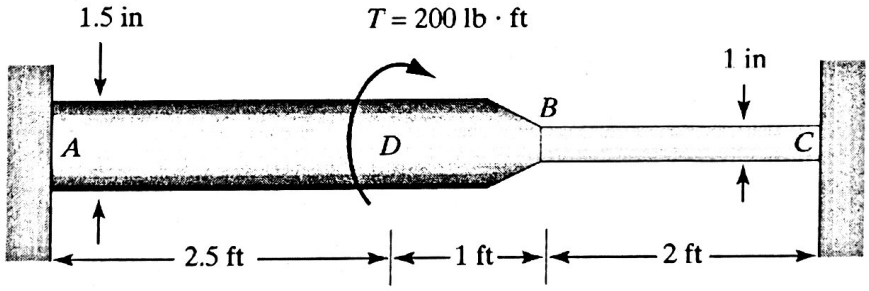
NOTICE: Write down the ANSYS commands on your answer sheet.

Problem 1

Consider a shaft that is made of two parts as shown in below figure. Part AB is made of material with a shear modulus of elasticity of $G_{AB} = 3.9 \times 10^6$ lb/in^2 and has a diameter of 1.5 in. Segment BC is made of slightly different material with a shear modulus of elasticity of $G_{BC} = 4.0 \times 10^6$ lb/in^2 and with a diameter of 1 in. The shaft is fixed at both ends. A torque of 200 lb·ft is applied at D. Assume that the elastic modulus of the material is $E = 10.4 \times 10^6$ lb/in^2 throughout the shaft.

(a) Determine the angle of twist at D and B.

(b) Obtain the torsional reactions at the boundaries.



GIF.

Problem 2

Consider a beam with a linear elastic spring attached to the middle of the beam, as shown in below figure. What is the minimum value of the spring constant k needed to ensure the nodal displacement that is 0.15m or less at position P of the beam? The beam is made of steel (E = 210 GPa) and A = 1×10^{-2} m^2, $I = 2\times10^{-4}$ m^4 throughout the beam. Use element type of LINK1 to model the steel spring which has a spring constant k.

